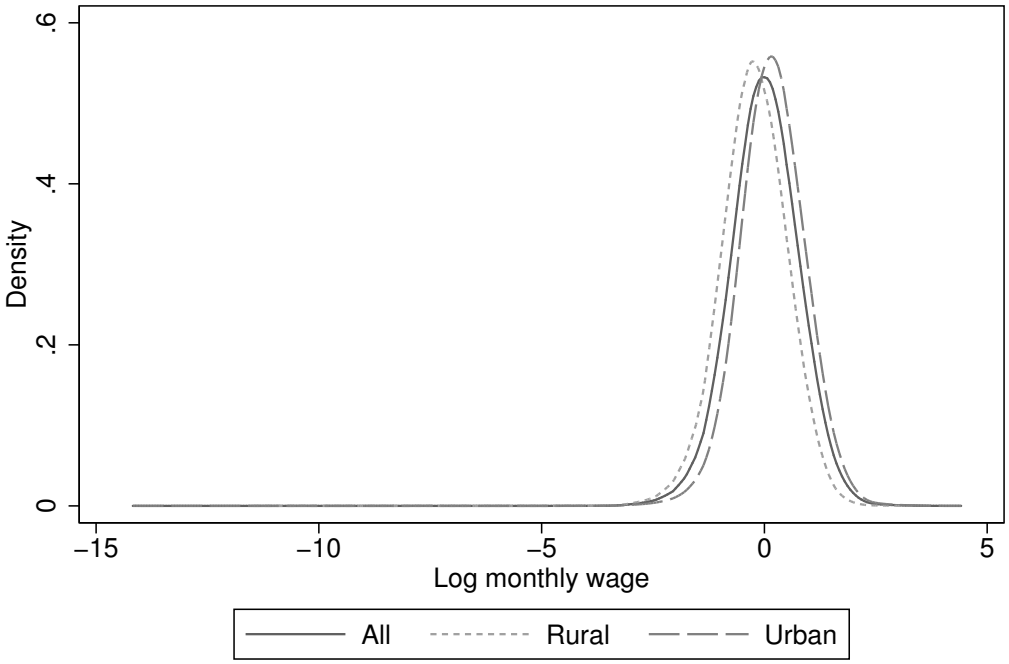
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Online Appendix

A Tables and Figures

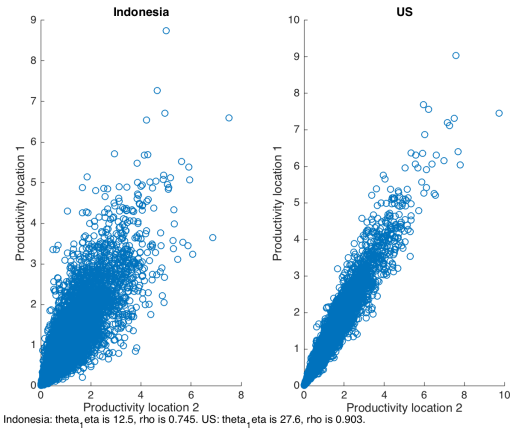
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Log monthly wage is demeaned of year fixed effects. Unit of observation is the individual.
Source: 1995 SUPAS, 2011 SUSENAS; 2012 SUSENAS.

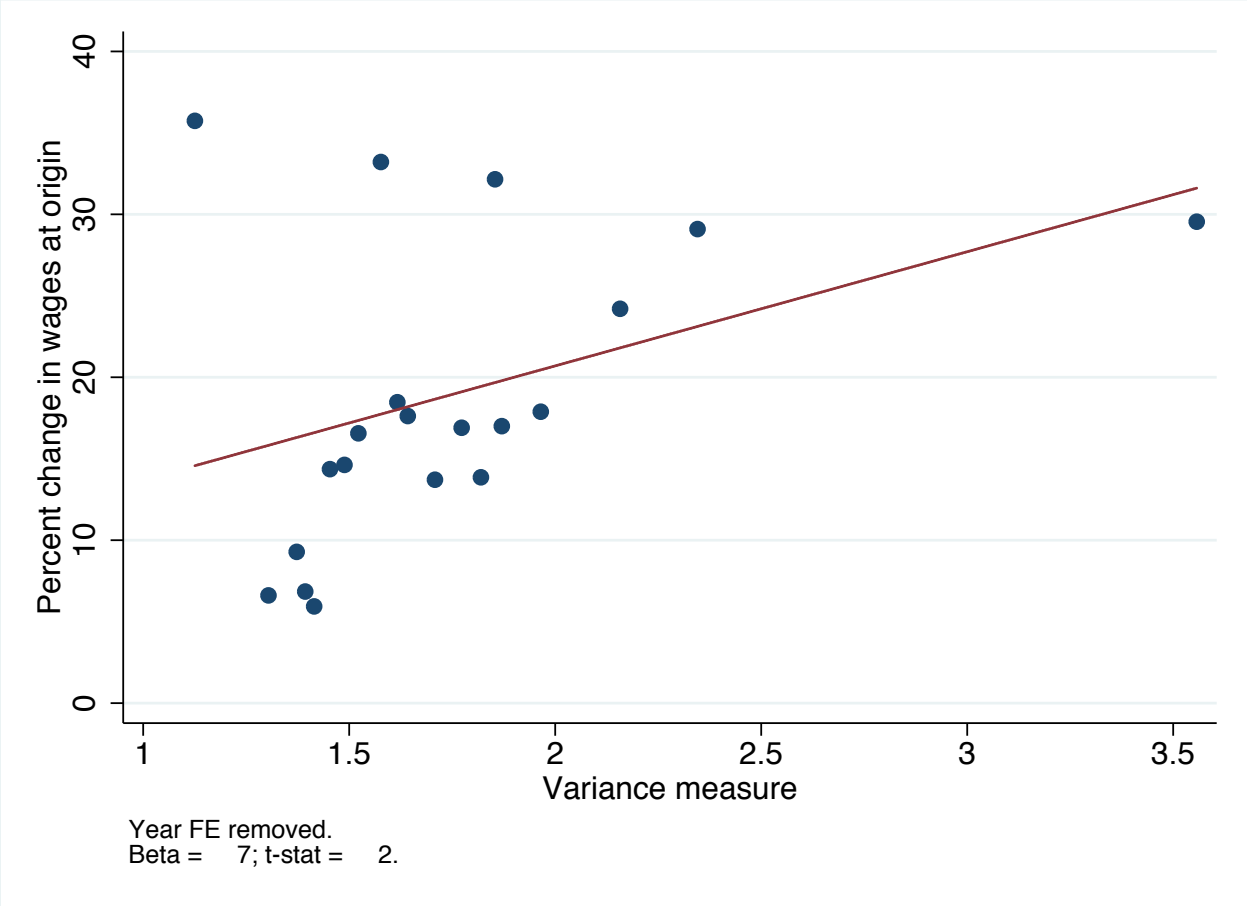
Appendix Figure 1: Distribution of wages: individual level

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Appendix Figure 2: Simulated Frechet Distribution

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Appendix Figure 3: Variance in origin wages

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Appendix Table 1: Summary statistics for Indonesia and US sample

	Indonesia			United States	
	(1)	(2)	(3)	(4)	(5)
	1995	2011	2012	1990	2010
Demographic					
Average age	38.58	39.63	39.86	40.78	44.30
Average age (mig)	38.43	39.31	39.71	41.26	44.95
Years school	8.01	9.85	10.12	13.48	15.24
Years school (mig)	10.17	11.12	11.33	14.03	15.61
High education	0.82	0.92	0.93	0.85	0.93
Financial					
Monthly non-zero wage	129.62	188.06	198.72	4424.07	4985.62
Monthly non-zero wage (mig)	174.82	236.25	246.36	4829.80	5667.14
GDP per capita	1328.48	3177.26	3337.69	39982.11	48377.39
Migration					
Share migrating	0.34	0.36	0.36	0.40	0.41
Share low educ migrate	0.15	0.19	0.19	0.31	0.30
Share high educ migrate	0.39	0.37	0.37	0.42	0.46
Share migrating (prov)	0.18	0.22	0.23		
Number of obs	59,006	68,510	69,330	1,958,123	336,033
Sum of sample weights	13,454,136	17,324,774	17,736,228	39,251,152	32,404,124

Notes: Sample is male head of household, between 15 and 65 years old. Migration defined at the regency level in Indonesia, and at the state level in the US. Low education is 3 years of schooling or less in Indonesia; 12 years or less in the US. Data source: Indonesia: 1995 SUPAS, 2011 SUSENAS, 2012 SUSENAS. US: 1990 ACS and 2010 ACS. GDP data from the World Bank Development Indicators Database. All financial values reported in 2010 USD.

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Appendix Table 2: Summary statistics for IFLS sample

	(1) 1993	(2) 1997	(3) 2000	(4) 2007
Demographic				
Average age	42.27	43.04	42.40	43.41
Average age (mig)	41.98	42.87	42.05	43.40
Years school	5.18	5.83	7.27	7.93
Years school (mig)	6.42	6.92	8.69	8.74
High education	0.73	0.79	0.83	0.87
Financial				
Monthly non-zero wage	94.05	111.51	114.82	128.11
Monthly non-zero wage (mig)	125.77	138.41	142.07	155.09
GDP per capita	1181.78	1552.12	1662.29	2276.16
Migration				
Share migrating	0.32	0.35	0.39	0.47
Share low educ migrate	0.22	0.23	0.27	0.35
Share high educ migrate	0.36	0.38	0.41	0.49
Share migrating (prov)	0.14	0.16	0.24	0.23
Number of obs	5,496	5,625	7,709	9,976
Sum of sample weights	5,501	5,952	8,430	10,810

Notes: Sample is male head of household, between 15 and 65 years old. Migration defined at the regency level. Low education is 3 years of schooling or less in Indonesia. Data source: 1993, 1997, 2000 and 2007 IFLS surveys. GDP data from the World Bank Development Indicators Database. All financial values reported in 2010 USD.

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Appendix Table 3: Migration histories

	Entire sample		Male HOH	
	Regency mean	Province mean	Regency mean	Province mean
Whole sample				
Current migrant	0.40	0.14	0.45	0.18
Ever migrated	0.45	0.18	0.48	0.21
People who have migrated at least once				
Migrate once	0.55	0.64	0.60	0.69
Migrate twice	0.32	0.30	0.29	0.26
Migrate three times or more	0.12	0.06	0.12	0.05
Number moves	1.61	1.44	1.56	1.38
Migrant who has returned home	0.12	0.19	0.08	0.14
First migration before age 16	0.16	0.17	0.05	0.08
Num. individuals	37612	37612	9425	9425

Notes: Data source: IFLS rounds 1-4. Entire sample is all adults in sample who report a birth location. Male HOH is people who are ever heads of households over the sample. Migration history collected for all moves after age 12.

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Appendix Table 4: Five facts about migration in Indonesia, province-level

	Movement costs		Selection		Compensating Diff.
	(1)	(2)	(3)	(4)	(5)
Dep. variable	$\log \pi_{odt}$	$\log w_{odt}$	$\log w_{odt}$	$\log w_{odt}$	$\log w_{odt}$
Log distance	-0.606	0.041		0.019	
	(0.013)***	(0.003)***		(0.009)**	
Log share migrating			-0.066	-0.036	
			(0.005)***	(0.012)***	
Amenities					-0.059
					(0.074)
Destination x Year FE	yes	yes	yes	yes	no
Origin x Year FE	yes	yes	yes	yes	yes
Destination FE					yes
No. of individuals	196846	196838	196838	196838	196838
No. of region pairs	1452	1444	1444	1444	1444

Notes: $\log \pi_{odt}$ is the log of the share of population migrating from o to d in year t. $\log w_{odt}$ is the log of the average wage of migrants from origin o in destination d in time t. An observation is an origin-destination province pair. Datasource: 1995 SUSENAS, 2011 SUSENAS, 2012 SUSENAS. Amenity measure is the standardized value of the first principal component. Two-way clustering of standard errors at the origin-year and destination-year reported in Columns (1)-(4). Clustered standard errors, at the level of the origin-year, reported in Column (5). Number of observations changes between columns because not all pairs with positive migration flows have observed wages.

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Appendix Table 5: Four facts about migration in Indonesia, IFLS data

	Movement costs		Selection	
	(1)	(2)	(3)	(4)
Dep. variable	$\log \pi_{odt}$	$\log w_{odt}$	$\log w_{odt}$	$\log w_{odt}$
All individuals				
Log distance	-0.571	0.041		-0.044
	(0.018)***	(0.006)***		(0.017)*
Log share migrating			-0.079	-0.150
			(0.010)***	(0.028)***
Destination x Year FE	yes	yes	yes	yes
Origin x Year FE	yes	yes	yes	yes
No. of individuals	26820	26732	26732	26732
No. of region pairs	613	546	546	546
Wage employees only				
Log distance	-0.527	0.026		0.000
	(0.015)***	(0.004)***		(0.013)
Log share migrating			-0.049	-0.049
			(0.008)***	(0.024)
Destination x Year FE	yes	yes	yes	yes
Origin x Year FE	yes	yes	yes	yes
No. of individuals	12241	12223	12223	12223
No. of region pairs	497	479	479	479

Notes: $\log \pi_{odt}$ is the log of the share of population migrating from o to d in year t. $\log w_{odt}$ is the log of the average wage of migrants from origin o in destination d in time t. An observation is an origin-destination province pair. Datasource: 1993, 1997, 2000, 2007 IFLS. Two-way clustering of standard errors at the origin-year and destination-year reported in Columns (1)-(4). Number of observations changes between columns because not all pairs with positive migration flows have observed wages.

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Appendix Table 6: Correlation of estimated amenities with data

	(1) 1995 b/se	(2) 2011 b/se	(3) 2012 b/se
Water pollution (past year)	-2.23***	-0.96**	-0.91**
	(0.86)	(0.42)	(0.38)
Land pollution (past year)	1.57	-5.15***	-4.91***
	(2.73)	(1.71)	(1.52)
Air pollution (past year)	-0.77***	0.16	-0.24
	(0.28)	(0.55)	(0.50)
Noise pollution (past year)	-2.46		
	(1.98)		
Main road village lighting	0.39	0.68*	0.68**
	(0.71)	(0.38)	(0.34)
Has movie theater	-10.6	-33.9	-77.1**
	(7.35)	(41.0)	(34.1)
Ease of reaching hospital	0.31**	0.10	0.27***
	(0.13)	(0.13)	(0.10)
Ease of reaching puskesmas/other health facility	0.44*	0.057	0.34**
	(0.26)	(0.20)	(0.17)
Ease of reaching market with permanent building	0.31**		
	(0.15)		
Ease of reaching shopping complex	0.37***		
	(0.13)		
Flooding		-0.46	-0.076
		(0.49)	(0.45)
Earthquake		-0.039	-0.073
		(0.21)	(0.19)
Whirlwind/tornado/hurricane		0.61	-0.20
		(0.41)	(0.38)
Drought		-0.52	0.53
		(1.20)	(1.09)
Outbreak (last year): Vomiting/diarrhea		-0.85	-0.81
		(0.72)	(0.65)
Outbreak (last year): Malaria		0.48	0.74**
		(0.42)	(0.36)
Outbreak (last year): Bird flu (1 case is considered an outbreak)		-10.2	-10.6
		(8.64)	(7.76)
Outbreak (last year): Tuberculosis		-0.59	-1.72*
		(1.13)	(0.97)

Notes: Data source: 1996 and 2011 PODES data and estimates from model. Table shows the regression coefficient between the estimated amenity value and the amenity measure given in each row. 1996 PODES data are correlated with the model estimates for 1995; 2011 PODES data are correlated with model estimates for 2011 and 2012.

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Appendix Table 7: Estimating spillover parameters

	Log wage		Log amenity		Log rents	
	(1) OLS	(2) IV	(3) OLS	(4) IV	(5) OLS	(6) IV
Log population	0.028	0.012	0.004	0.042	0.228	0.252
	(0.083)	(0.210)	(0.085)	(0.215)	(0.098)**	(0.247)
N	75	75	75	75	75	75
r2	0.89	0.89	0.76	0.76	0.99	0.99
F	0.11	0.00	0.00	0.04	5.38	1.04

Notes: Table shows OLS and 2SLS estimates using the model-based instrument for labor to estimate the strength of the agglomeration and congestion parameters. Province and year FE included in all specifications.

Appendix Table 8: Robustness: effect of migration cost on growth, Indonesia

	(1) Sub. elasticity = 4	(2) Sub. elasticity= 6	(3) Sub. elasticity = 8
Productivity spillover = 0			
$\lambda = 0$	1.113	1.107	1.098
$\lambda = .02$	1.112	1.102	1.086
$\lambda = -.02$	1.114	1.110	1.104
Productivity spillover = 0.05			
$\lambda = 0$	1.117	1.103	1.075
$\lambda = .02$	1.115	1.090	1.032
$\lambda = -.02$	1.118	1.110	1.096
Productivity spillover = 0.08			
$\lambda = 0$	1.119	1.094	1.033
$\lambda = .02$	1.115	1.069	0.929
$\lambda = -.02$	1.121	1.107	1.078

Notes: Table shows the effect of reducing migration cost to zero on labor output. Table shows different combinations of amenity (λ) and productivity spillovers, for different values of substitution parameter. Calculated for model with selection.

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Appendix Table 9: Robustness: effect of migration cost and housing on growth, Indonesia

	(1) Sub. elasticity = 4	(2) Sub. elasticity= 6	(3) Sub. elasticity = 8
Productivity spillover = 0			
$\lambda = 0$	1.115	1.114	1.112
$\lambda = .02$	1.115	1.113	1.111
$\lambda = -.02$	1.115	1.114	1.113
Productivity spillover = 0.05			
$\lambda = 0$	1.121	1.118	1.116
$\lambda = .02$	1.120	1.117	1.112
$\lambda = -.02$	1.121	1.119	1.118
Productivity spillover = 0.08			
$\lambda = 0$	1.124	1.121	1.117
$\lambda = .02$	1.123	1.118	1.111
$\lambda = -.02$	1.124	1.122	1.120

Notes: Table shows the effect of reducing migration cost and housing to zero on labor output. Table shows different combinations of amenity (λ) and productivity spillovers, for different values of substitution parameter. Calculated for model with selection.

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Appendix Table 10: Robustness: effect of amenities on growth, Indonesia

	(1) Sub. elasticity = 4	(2) Sub. elasticity= 6	(3) Sub. elasticity = 8
Productivity spillover = 0			
$\lambda = 0$	1.063	1.083	1.098
$\lambda = .02$	1.066	1.087	1.103
$\lambda = -.02$	1.059	1.079	1.093
Productivity spillover = 0.05			
$\lambda = 0$	1.074	1.103	1.127
$\lambda = .02$	1.078	1.108	1.133
$\lambda = -.02$	1.070	1.098	1.121
Productivity spillover = 0.08			
$\lambda = 0$	1.082	1.119	1.152
$\lambda = .02$	1.086	1.125	1.158
$\lambda = -.02$	1.078	1.114	1.145

Notes: Table shows the effect of reducing amenities to zero on labor output. Table shows different combinations of amenity (λ) and productivity spillovers, for different values of substitution parameter. Calculated for model with selection.

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Appendix Table 11: Robustness: effect of migration cost and amenities on growth, Indonesia

	(1) Sub. elasticity = 4	(2) Sub. elasticity= 6	(3) Sub. elasticity = 8
Productivity spillover = 0			
$\lambda = 0$	1.163	1.177	1.189
$\lambda = .02$	1.166	1.182	1.193
$\lambda = -.02$	1.159	1.173	1.184
Productivity spillover = 0.05			
$\lambda = 0$	1.177	1.199	1.217
$\lambda = .02$	1.182	1.204	1.223
$\lambda = -.02$	1.173	1.194	1.211
Productivity spillover = 0.08			
$\lambda = 0$	1.187	1.215	1.240
$\lambda = .02$	1.192	1.221	1.247
$\lambda = -.02$	1.183	1.209	1.233

Notes: Table shows the effect of reducing migration cost and amenities to zero on labor output. Table shows different combinations of amenity (λ) and productivity spillovers, for different values of substitution parameter. Calculated for model with selection.

Appendix Table 12: Output gain from reducing migration barriers: robustness over theta

	(1) Mig costs	(2) Amenities	(3) Mig costs, amenities
Estimated θ			
Baseline	1.075	1.127	1.217
No selection	0.914	1.127	1.133
Set $\hat{\theta} = 0.5\theta$			
Baseline	1.115	1.120	1.231
No selection	0.923	1.120	1.127

Notes: Table shows the output gain from removing the barrier completely. Data is 1995, 2011, 2012 for Indonesia. No selection recalculates the output gain shutting down the role for comparative advantage.

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Appendix Table 13: Output gain from reducing migration barriers: robustness

	(1) Mig costs	(2) Amenities	(3) Mig costs, amenities
Baseline			
Baseline	1.075	1.127	1.217
No selection	0.914	1.127	1.133
Migration costs parameteric fn. of distance			
Baseline	1.094	1.122	1.219
No selection	0.935	1.112	1.131
Migration costs parameteric fn. of distance, lg pop interaction			
Baseline	1.170	1.146	1.303
No selection	0.949	1.108	1.156

Notes: Table shows the output gain from removing the barrier completely. Data is 1995, 2011, 2012 for Indonesia. No selection recalculates the output gain shutting down the role for comparative advantage.

Appendix Table 14: Output gain from reducing migration barriers: robustness over education

	(1) Mig costs	(2) Amenities	(3) Mig costs, amenities
Baseline (education fully adjusts)			
Baseline	1.075	1.127	1.217
No selection	0.914	1.127	1.133
Not allowing education to adjust			
Baseline	1.045	1.095	1.183
No selection	0.919	1.128	1.137

Notes: Table shows the output gain from removing the barrier completely. Data is 1995, 2011, 2012 for Indonesia. No selection recalculates the output gain shutting down the role for comparative advantage.

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B Model Extensions and Interpretation

This section discusses several extensions to the model: dynamics; endogenous human capital; non-traded goods; and costly trade.

B.1 Dynamics

We aim to understand whether improving the static allocation of workers would lead to increased productivity. Consequently, we rule out potential dynamic gains and leave quantifying these for future work. Some dynamic issues, however, could be a source of bias in our estimates, and we discuss those in this section.

To understand how to map our model to a dynamic setting, suppose that time is discrete and indexed by t. Our preferred interpretation is as follows. In each period t, every individual observes current wages, movement costs, and amenities, and decides where to live for the next period. She moves, and her labor for period t is consumed. The same process occurs again in period $t+1$, and on forever. If the parameters of the model are fairly stable, most people will move only one time in their working life. This interpretation can be linked to static trade models if one considers the migration decision as the choice to move to the destination for one period of work.

Given this interpretation, our estimates may be biased if individuals do not maximize their static amenity-adjusted wages, net of movement costs. One possibility is the channel emphasized in Caliendo et al. (2017). They model workers who face a cost of moving away from their current location and therefore chose the destination to maximize a combination of static values and the option value of future moves. Dynamic investment in human capital raises a separate but similar concern. People may migrate to a specific location to invest in human capital and be paid temporarily below their market wage (e.g., going to grad school). Alternatively, some dynamic choices may lead to temporary migration for high short-term wages (e.g., an unpleasant seasonal fishing job in Alaska to raise money for grad school). If choices of these types are prevalent, we will record the incorrect wages for these individuals, potentially biasing our estimates.

Based on the IFLS migration histories, discussed in Section 2.1 and Appendix Table 3 we concluded that migration in Indonesia is best characterized by one migration event, made in adulthood, and hence that these dynamic concerns are unlikely to be a substantial source of bias.

B.2 Endogenous Human Capital

Our base model treats education in a particularly simple way. All people from a particular origin receive the same education q_o, this level of education is exogenous and independent of movement frictions, and education is equally valuable in all destinations. This set of assumptions may bias our estimates of important parameters (for example, θ the selection parameter) and/or affect our counterfactuals. We discuss each possibility in turn.

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One way to understand biases is to experiment with different ways of incorporating education. One approach is to assume that

$$h_{id} = s_{id} q_o e^{\eta},$$

where e is the chosen level of education and η measures the elasticity of human capital (and hence wage) to a change in education. If we combine this production function with the strong assumption that workers make their human capital decisions with the aim of maximizing wages, rather than utility, solving

$$\max_e \left(w_d e_{do}^w s_{id} q_o e^{\eta} - \eta e \right),$$

then equation (6) becomes

$$\overline{wage}_{do} = w_d^{\frac{1}{1-\eta}} \epsilon_{do}^w q_o^{\frac{1}{1-\eta}} \pi_{do}^{-\frac{1}{\theta(1-\eta)}} \bar{\Gamma}.$$

This way of modelling education preserves the key identification requirement (discussed above) that π_{do} is sufficient to summarize the impact of amenities and movement costs on average wages. Hence, identification of the model proceeds as as in the main text, with slightly different interpretations of the parameters. For example, suppose that differences in movement costs lead to $\pi_{do} > \pi_{d'o}$. In the extended model with endogenous human capital, this will lead to $\overline{wage}_{do} < \overline{wage}_{d'o}$ for two reasons. First, as in the baseline model, those that move to d' are more selected. Second, because those who move to d' are more selected, they are also more educated. That is, education creates a complementarity which increases the impact of selection. Below we concentrate our efforts on estimating the elasticity of average wage to π_{do}, if education is not endogenous (as in the base model) this elasticity captures only the selection effect, if education is endogenous it captures a composite effect. That is, endogenous human capital modeled in this way does not create bias, but does change interpretation.

If we relax the assumption that education is chosen to maximize wage, assuming instead that workers try to maximize utility, we no longer maintain the structure that τ_{do} and α_d enter the wage equation only through π_{do}. This would mean that our method of identification is no longer valid. Fortunately, we have already presented some evidence that π_{do} is sufficient. Fact 4 above shows that, after controlling for π_{do}, distance (our proxy for movement costs) no longer correlates with $\overline{wage}_{do}$. Given this evidence, we conclude that endogenous education is unlikely to be a major source of bias in our estimates, noting only that the interpretation of the estimated θ is different if education choice is added to the model.

The model incorporating endogenous education also highlights a concern for our counterfactuals. If dynamics are thought of as discussed in Section B.1, then it becomes clear that we are assuming that all individuals, even the elderly, are able to optimize their human capital choices in response to changes in movement costs. In the model presented above, human capital investment will increase as movement costs decrease because workers are better able to match to their comparative advantage and human capital is complementary to skill. This issue is difficult to deal with while retaining the structure

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of the model. We address this issue quantitatively in a discussion in Section 6.5, using the model of endogenous human capital presented above. In this section, we set the value of η to zero and recompute the model. To do this, we need an initial estimate of η. We use the approach in Hsieh et al. (2018) and calculate this by dividing the share of GDP spent on education in Indonesia, which is 2.4% (The World Bank, 2014) by the share of labor income in GDP, 0.45 (Heston et al., 2006). This yields $\eta = 0.05$.[27]

B.3 Non-traded Goods

Prices of nontradeables, particularly housing, are likely to be an endogenous function of population. In our baseline model, nontradeable prices will be captured in the amenity term, α_d, and λ, which measures the extent to which amenities change with population. When we discuss calibration of λ below, we will be particularly concerned about endogenous house prices, so we briefly outline here how α_d can be decomposed into rents and other amenities.

Following Monte et al. (2018), let utility be a Cobb-Douglas function of the consumption aggregate and housing, with β_h measuring the share of expenditure on housing. The indirect utility function (3) then becomes

$$U_{ido} = \frac{\tilde{\alpha}_d}{r_d^{\beta_h}} \epsilon_{do}^{\alpha} (1 - \tau_{do}) w_d \epsilon_{do}^{w} s_{id} q_o \equiv \bar{w}_{do} s_{id}$$

where r_d is the price of housing in location d and $\tilde{\alpha}_d$ is a measure of all other amenities in that location. When we estimate amenities below, we will recover $\log \alpha_d = \log \tilde{\alpha}_d - \beta_h \log r_d$. The endogenous effect of population on α_d can also be decomposed. If $\tilde{\alpha}_d = \overline{\alpha}_d \hat{L}_d^{\lambda_a}$ and $r_d = \hat{L}_d^{\lambda_r}$ then $\alpha_d = \overline{\alpha}_d \hat{L}_d^{\lambda_\alpha - \beta_h \lambda_r}$. We separately calibrate λ_α and λ_r when undertaking counterfactuals in Section 6.

B.4 Costly Trade

Goods trade within Indonesia is surely costly, and incorporating trade costs could change our results. The main source of concern is not with estimation – differences in consumer goods prices caused by costly trade will be captured in the amenity term – but with counterfactuals, which may change if endogenous changes in the price of traded goods were accounted for.

There are two main sources of interaction. First, many policies that decrease movement costs would also decrease trade costs. For example, road widening would likely decrease the cost of moving goods and also decrease the amount of time required to visit relatives. We specifically *do not* want to capture this interaction between trade and migration costs. Our purpose in writing the paper is to understand whether migration frictions are an independent cause of low productivity.

Second, changes in movement costs will interact with (trade) market access. Burstein et al. (2017) discusses one example. They show that an industry's ability to accept new

[27] Hsieh et al. (2018) find a value of 0.1 for the U.S. This comes from an expenditure share on education of 6.6%, divided by the share of labor income in GDP of 0.64.

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labor without price (or wage) decreases depends on trade costs. Incorporating this channel would add an additional dimension of heterogeneity across destinations. While we do not model this channel, σ (the elasticity of substitution) limits the extent to which one destination can accept migrants, and acts in a similar manner. A second interaction occurs if reducing migration frictions leads to an increase in productivity of one destination d, with a knock-on effect in other destinations that have greater access to d, which now produces lower priced goods. We note that our counterfactuals should be interpreted with this caveat.

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C IV Approach and Monte Carlo Simulation

This appendix illustrates the IV approach discussed in Section 4. The estimating equation, equation (7), is:

$$\ln(\overline{wage}_{do}) = \underbrace{\ln(\overline{\Gamma}) + \ln(w_d)}_{\text{Destination fixed effect}} - \frac{1}{\theta}\ln(\pi_{do}) + \underbrace{\ln(q_o)}_{\text{Origin fixed effect}} + \ln \epsilon^w_{do}, \quad (10)$$

where the migration rate, π_{do}, is defined in equation (4) as:

$$\pi_{do} = \frac{w_d \epsilon^w_{do} \alpha_d \epsilon^\alpha_{do} (1 - \tau_{do})}{\sum_{j=1}^{J} w_j \epsilon^w_{jo} \alpha_j \epsilon^\alpha_{jo} (1 - \tau_{jo})}.$$

There is an endogeneity problem because the shock, ϵ^w_{do} appears in both the migration rate and the wage. As a result, OLS estimates are biased, with the implication that θ will be overestimated.

We propose an IV strategy that uses the common utility of the destination, estimated using shares from all other origins o', to provide a "pull" factor for the destination. This approach has links to the IV estimation of demand systems in IO, such as that in Berry (1994). We do the following three steps:

1. We construct a vector z^1_{od} consisting of all the flows and squared flows to a destination (dropping the dependent variable flows π_{od}): $\{\log \pi_{1d}, ..., \log \pi_{jd}, \log \pi^2_{1d}, ..., \log \pi^2_{jd}\}\forall j \neq o$. We then run a "zero stage" regression and predict the fitted values $\widehat{\log \pi}_{od} = \hat{\beta} z^1_{od}$ from a linear regression.

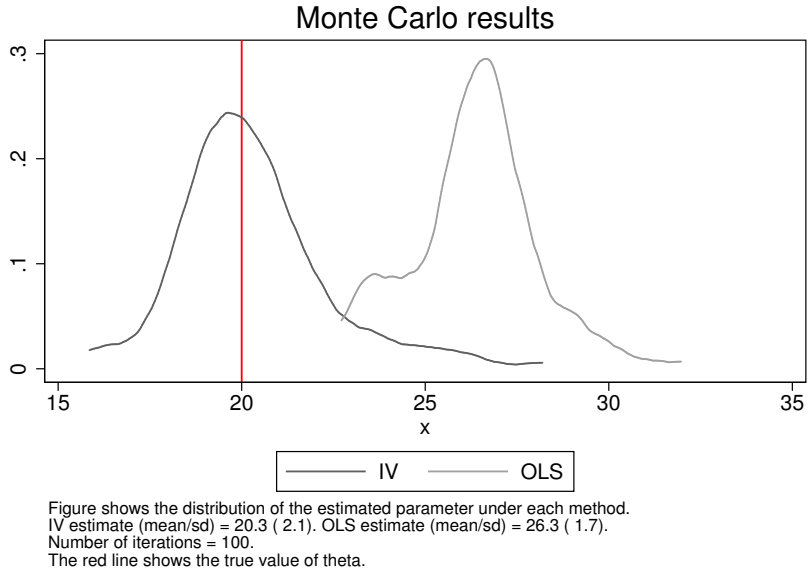
2. We then run the first stage regression, where we regress $\log \pi_{ij}$ on $\widehat{\log \pi}_{ij}$ and a full set of origin-year and destination-year fixed effects.

3. We then run the second stage regression, where we regress $\log \text{wage}_{ij}$ on $\log \pi_{ij}$, instrumenting for $\log \pi_{ij}$ with $\widehat{\log \pi}_{ij}$ and a full set of origin-year and destination-year fixed effects, and then controlling for the full set of origin-year and destination-year FE.

This approach follows Angrist and Pischke (2009).

Appendix Figure 4 shows the results of a simulation of our model. As predicted, the OLS estimate of θ is upward bias. The IV estimator corrects this bias and is not statistically different than the true value. Appendix Table 16 shows that our IV estimate of θ is robust to several different definitions of the shocks used in the construction of the instrument.

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Appendix Figure 4: Monte Carlo Simulations

Appendix Table 15: First stage regression for migration flows

Dep var: Log prob	(1) Baseline	(2) Drop large	(3) Drop close
Predicted log probability	9.151	6.340	7.140
	(0.241)***	(0.232)***	(0.244)***
Destination X Year FE	yes	yes	yes
Origin X Year FE	yes	yes	yes
N	1452	1452	1452
r2	0.75	0.66	0.68
F	1446.16	748.63	858.38

Notes: Table shows first-stage regression of migration probability on predicted probability. The unit of observation is a province-to-province-year flow. Column (2) drops the largest 5% of flows when constructing the instrument. Column (3) drops observations within 500 km of a location when constructing the instrument.

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Appendix Table 16: Robustness: alternative instruments

	(1) Baseline	(2) Dropping large regions	(3) Dropping close regions
θ (dispersion)	12.2***	14.1***	13.5***
	(1.26)	(2.00)	(1.81)

Notes: Table shows the estimate of theta from the baseline IV specification in Columnn (1). Column (2) drops the top 5% of observations when constructing the instrument. Column (3) drops locations closer than 500km when constructing the instrument. Bootstrapped standard errors reported.

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D Can the Data Reject our Distributional Assumptions?

Our assumption of a multivariate Fréchet (Gumbel copula with Fréchet marginals) has important advantages. In particular, the distribution is closed under maximization, and so we get closed form solutions and transparent identification of key parameters. We consider this transparent identification one of the key contributions of the paper. Other copulas, such as the Frank copula considered by Lagakos and Waugh (2013), are not closed under maximization and therefore do not lead to closed form solution.

Nevertheless, it is important to explore whether the data supports the strong structure imposed by our distributional choice. The multivariate Fréchet implies two important assumptions. First, absolute and comparative advantage are aligned.[28] This implies that removing barriers to migration will reduce the difference between average wages of people born in the same origin, and living in different destinations. Second, the Fréchet distribution, combined with the assumption that the distributional parameter θ does not differ by origin, implies that average wages are a constant elasticity function of the proportion of people who migrate. This in turn implies that removing all barriers to migration will lead to complete convergence of wages within origin, with only the origin fixed effect q_o leading to variation across origins (and will lead to complete convergence in wages per unit of human capital across origins). Hence, selection alone cannot generate average wage differences in our model. This is in strong distinction to the model presented in Young (2013).

We first look for evidence supporting the assumption that comparative and absolute advantage are aligned. Young (2014) argues that if, on average, the elasticity of the average wage with respect to the proportion of the population in that sector is negative, then comparative advantage is on average aligned with absolute advantage. This is shown in Table 2: the elasticity of average wage to proportion migrating is negative, hence θ (the negative inverse of the elasticity) is positive. As pointed out by Young (2014), it is probably inappropriate to disaggregate these measures too much, Nevertheless, we present results broken down by destination and origin in Appendix Table 17. These results show that, although the estimates are very noisy, in all but 5 cases in Column (2) (6 cases for Column (3)), the elasticity is positive. So we feel comfortable concluding that comparative advantage is aligned with absolute advantage in the data.

The second assumption is that the elasticity of wages to population shares is constant across all locations. The above exercise implies that we cannot reject that the shape parameters are the same, although the test has little power. A second implication is that, in the absence of amenity differences, the wage gap should be equal to zero at when movement costs are zero. One way to empirically test this is to look at whether wage differences drop to zero as distance (a proxy for movement costs) drops to zero. Appendix Table 18 below shows a regression of the log wage gap on the log distance. (Note that we do not include amenities here because they will on average cancel out due to symmetry.)

[28] To see this, note that the average skill, conditional on selecting into location d, is given by $E(\epsilon_d|\text{choose } d) = \Gamma\pi_{od}^{\frac{-1}{\theta}}$. The unconditional mean of the Frechet distribution is given by $E(\epsilon_d) = \Gamma$. Therefore, $\frac{E(\epsilon_d|\text{choose } d)}{E(\epsilon_d)} = \pi_{od}^{\frac{-1}{\theta}}$ which is always greater than 1, because $\theta \geq 2$ in order for the variance of the Frechet distribution to be defined.

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The intercept is not statistically significant, although imprecisely estimated. That is, we see no evidence of the presence of a wage gap at a hypothetical zero distance.

Taken together, we do not find any empirical evidence to suggest that our empirical specification fails to match the data. There is no strong evidence to suggest that the distributional assumption we make would lead to incorrect inference on either the amenities or movement costs.

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Appendix Table 17: Robustness: does theta vary by destination or origin?

	(1) Pooled	(2) Destination-specific	(3) Origin-specific
θ (dispersion)	12.2***		
	(1.26)		
θ, region 1		20.8	5.48
		(96.8)	(141.9)
θ, region 2		17.7	6.11
		(71.2)	(25.6)
θ, region 3		-103.7	6.23
		(981.1)	(49.4)
θ, region 4		-5.31	25.8
		(83.3)	(67.4)
θ, region 5		22.3	11.9
		(72.1)	(78.8)
θ, region 6		8.18	-7.15
		(12.6)	(80.2)
θ, region 7		-2.82	4.85
		(83.2)	(120.2)
θ, region 8		45.6	2.02
		(375.8)	(18.9)
θ, region 9		16.4	8.90
		(107.5)	(24.0)
θ, region 10		-21.6	9.41
		(134.2)	(14.0)
θ, region 11		7.37	46.9
		(26.2)	(338.2)
θ, region 12		9.62	0.023
		(18.5)	(65.6)
θ, region 13		17.4	-34537.4
		(82.9)	(345255.5)
θ, region 14		27.2	-302.4
		(267.1)	(3053.5)
θ, region 15		8.63*	10.9
		(4.43)	(98.2)
θ, region 16		10.6	3.46
		(61.2)	(31.3)
θ, region 17		12.9	678.1
		(77.2)	(6746.0)
θ, region 18		-2.38	99.2
		(57.3)	(894.9)
θ, region 19		8.14	26.4
		(9.98)	(163.7)
θ, region 20		4.79	-24.5
		(4.09)	(122.4)
θ, region 21		15.4	15.0
		(42.1)	(183.8)
θ, region 22		5.01	-13.8
		(92.5)	(77.2)
θ, region 23		8.79	3.23
		(22.1)	(19.9)
θ, region 24		80.9	-6.93
		(740.0)	(15.5)
θ, region 25		12.6	2.95
		(40.7)	(47.2)
P value from F-test		1.00	1.00

Notes: Table shows the estimate of theta from the pooled data, and then the specification where this it is estimated region-by-region. The p value is the p value from the F test that all coefficients are equal. IV estimates.

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Appendix Table 18: Wage ratio and distance

Dependent variable: $\log \frac{wage_do}{wage_oo}$	(1) b/se	(2) b/se
Log distance +1	0.032	0.032
	(0.018)*	(0.018)*
Constant	0.042	0.067
	(0.135)	(0.136)
Year FE	No	Yes

Notes: Table testing hypothesis that the wage gap is zero when transport costs are zero.